EXERCISE 7

The Nilpotent Case: Zero as a Repeated Eigenvalue

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Nilpotent Matrix • Shear • Line of Equilibria

STATEMENT

Suppose that a non-zero matrix A has zero as a repeated eigenvalue, with eigenvector $\mathbf{v}$. Then we can change to coordinates referred to $\mathbf{v}_2$ and $\mathbf{v}_1 = A\mathbf{v}_2$, where $\mathbf{v}_2$ is any vector in a different direction to $\mathbf{v}$, so that $\tilde{\mathbf{x}} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$. The equation $d\mathbf{x}/dt = A\mathbf{x}$ becomes $\dfrac{d\tilde{\mathbf{x}}}{dt} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \tilde{\mathbf{x}}$, hence,

$$\frac{d\tilde{x}}{dt} = \tilde{y} \quad \text{and} \quad \frac{d\tilde{y}}{dt} = 0.$$

- (i) Solve the equations for $\tilde{x}(t)$ and $\tilde{y}(t)$, and hence write down the general solution for $\mathbf{x}(t)$.
- (ii) Draw the phase diagram in the $(\tilde{x}, \tilde{y})$ and (x, y) planes.
- (iii) Draw the phase diagram for the equation $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \mathbf{x}$.

PHASE 1: SETUP AND HYPOTHESES

$A \neq O$ is 2×2 with the double eigenvalue 0: $\operatorname{tr} A = 0$, $\det A = 0$, so by Cayley-Hamilton $A^2 = O$ ($A^2 - (\operatorname{tr} A)A + (\det A)I = O$). Since $A \neq O$, $\ker A$ is one-dimensional, spanned by $\mathbf{v}$: $m_g = 1$ (Topic 4 Sec. 8.4 with $\lambda = 0$, $N = A$).

Why the proposed basis works: take $\mathbf{v}_2 \notin \operatorname{span}\{\mathbf{v}\}$. Then $A\mathbf{v}_2 \neq \mathbf{0}$ (otherwise $\mathbf{v}_2$ would be in $\ker A$), and $\mathbf{v}_1 = A\mathbf{v}_2$ satisfies $A\mathbf{v}_1 = A^2\mathbf{v}_2 = \mathbf{0}$, so $\mathbf{v}_1 \in \ker A = \operatorname{span}\{\mathbf{v}\}$ and $\mathbf{v}_1 \neq \mathbf{0}$ is an eigenvector. $\mathbf{v}_1, \mathbf{v}_2$ are independent (not parallel), and

$$A\mathbf{v}_1 = \mathbf{0} = 0\cdot\mathbf{v}_1 + 0\cdot\mathbf{v}_2, \qquad A\mathbf{v}_2 = \mathbf{v}_1 = 1\cdot\mathbf{v}_1 + 0\cdot\mathbf{v}_2 \implies Q^{-1}AQ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

PHASE 2: FORMULATION

With $Q = [\mathbf{v}_1 \mid \mathbf{v}_2]$ and $\mathbf{x} = Q\tilde{\mathbf{x}} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$ the canonical system is $\dot{\tilde{x}} = \tilde{y}, \dot{\tilde{y}} = 0$ (the Jordan block of Topic 4 Eq. 8.3 with $\lambda = 0$).

PHASE 3: DERIVATION

(i) Solution

The second equation $\dot{\tilde{y}} = 0$ integrates to $\tilde{y}(t) = c_2$ (a constant). Substituting in the first, $\dot{\tilde{x}} = c_2$, and Barrow's rule between 0 and t gives $\tilde{x}(t) - \tilde{x}(0) = \int_0^t c_2\, ds = c_2 t$:

$$\tilde{x}(t) = c_1 + c_2 t, \qquad \tilde{y}(t) = c_2 \qquad (c_1 = \tilde{x}(0),\ c_2 = \tilde{y}(0))$$

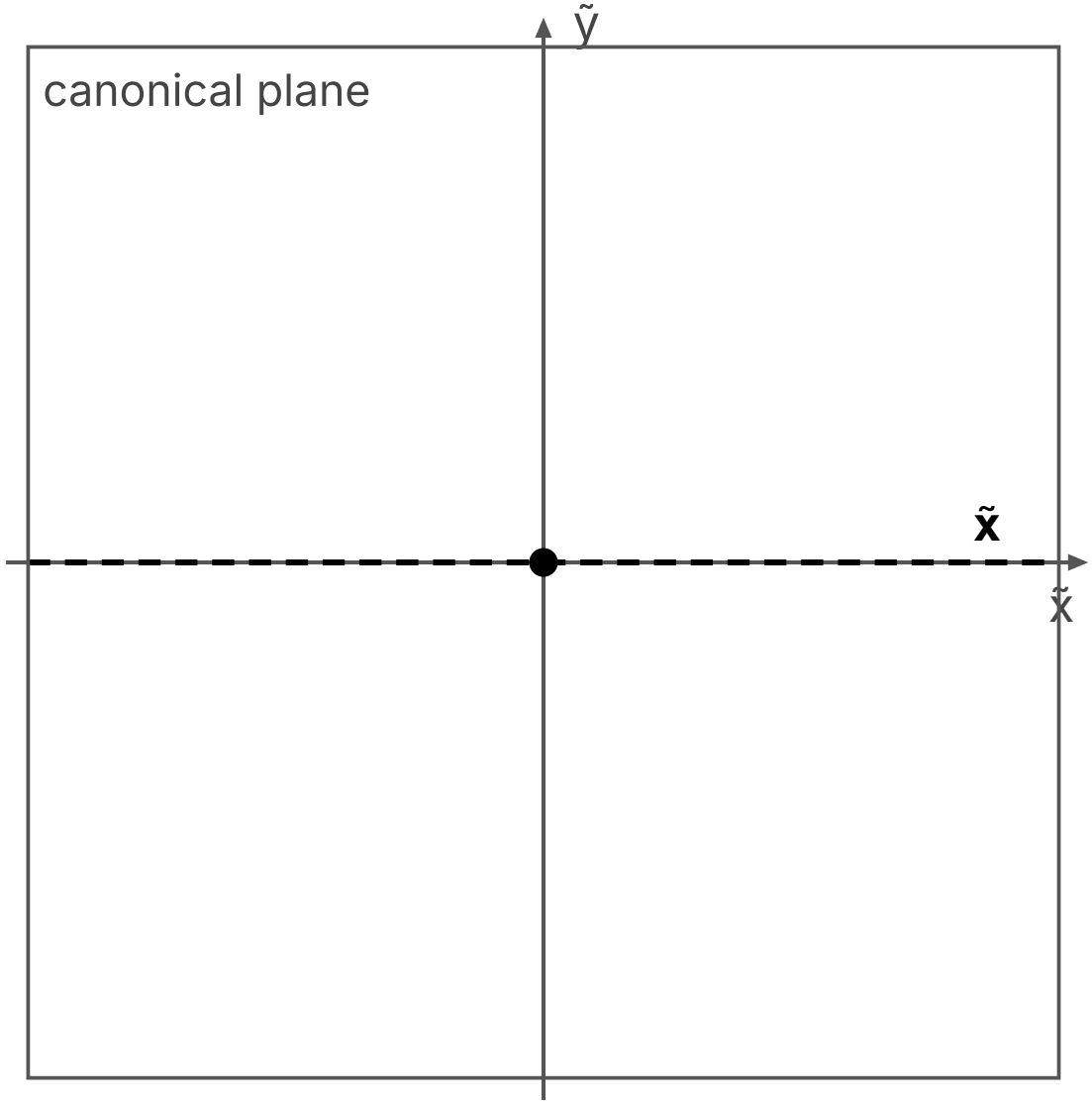
Back to the original coordinates, $\mathbf{x} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$:

$$\boxed{\mathbf{x}(t) = (c_1 + c_2 t)\,\mathbf{v}_1 + c_2\,\mathbf{v}_2}$$

Check: $\dot{\mathbf{x}} = c_2\mathbf{v}_1$ and $A\mathbf{x} = (c_1 + c_2 t)A\mathbf{v}_1 + c_2 A\mathbf{v}_2 = \mathbf{0} + c_2\mathbf{v}_1$. They agree. Unlike the case $\lambda \neq 0$ there is no exponential: the secular term grows only **linearly**.

(ii) Phase diagrams

In the $(\tilde{x}, \tilde{y})$ plane the orbits are the **horizontal lines** $\tilde{y} = c_2$ (since $\tilde{y}$ is constant). On the line $\tilde{y} = c_2 > 0$ the abscissa $\tilde{x} = c_1 + c_2 t$ increases at constant speed c_2: motion to the right; for $c_2 < 0$ to the left. The $\tilde{x}$-axis ($c_2 = 0$) consists of equilibria. There are no curved orbits: the speed is proportional to the height $\tilde{y}$ (a *shear* flow).



$(\tilde{x}, \tilde{y})$-plane: horizontal lines $\tilde{y} = c_2$ traversed to the right above the axis and to the left below it; the $\tilde{x}$-axis is all equilibria.

In the (x, y) plane, $\mathbf{x} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$ maps horizontal lines to **lines parallel to $\mathbf{v}_1$**, offset from the line $\operatorname{span}\{\mathbf{v}_1\}$ of equilibria by $c_2\mathbf{v}_2$; they are traversed along $+\mathbf{v}_1$ on the side of $\operatorname{span}\{\mathbf{v}_1\}$ where $\mathbf{v}_2$ points and along $-\mathbf{v}_1$ on the other side. The (x, y)-diagram of the following item shows a concrete example.

(iii) The matrix $\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}$

$\operatorname{tr} A = -1 + 1 = 0$ and $\det A = (-1)(1) - (1)(-1) = 0$, so the characteristic equation is $\lambda^2 = 0$: the double eigenvalue 0, and $A \neq O$. Check $A^2 = O$:

$$
\begin{aligned}
(A^2)_{11} &= (-1)\cdot(-1) + 1\cdot(-1) = 0 \\
(A^2)_{12} &= (-1)\cdot 1 + 1\cdot 1 = 0 \\
(A^2)_{21} &= (-1)\cdot(-1) + 1\cdot(-1) = 0 \\
(A^2)_{22} &= (-1)\cdot 1 + 1\cdot 1 = 0
\end{aligned}
$$

Eigenvector: $A\mathbf{v} = \mathbf{0}$ gives $-v_1 + v_2 = 0$ (both rows are the same equation up to sign), so $\mathbf{v} = (1,1)^T$. Following the exercise, take any $\mathbf{v}_2$ not parallel to $\mathbf{v}$, for instance $\mathbf{v}_2 = (1,0)^T$, and set $\mathbf{v}_1 = A\mathbf{v}_2$:

$$
\mathbf{v}_1 = A\mathbf{v}_2 = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}
$$

$\mathbf{v}_1 = (-1,-1)^T$ is indeed an eigenvector ($A\mathbf{v}_1 = (1-1, 1-1)^T = \mathbf{0}$), a multiple of $\mathbf{v}$.

$$
Q = [\mathbf{v}_1 \mid \mathbf{v}_2] = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}
$$

$\det Q = (-1)(0) - (1)(-1) = 1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$
Q^{-1} = \frac{1}{1}\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}
$$

$$
AQ = [A\mathbf{v}_1 \mid A\mathbf{v}_2] = [\mathbf{0} \mid \mathbf{v}_1] = \begin{pmatrix} 0 & -1 \\ 0 & -1 \end{pmatrix}
$$

$$
\begin{aligned}
(Q^{-1}AQ)_{11} &= 0\cdot 0 + (-1)\cdot 0 = 0 \\
(Q^{-1}AQ)_{12} &= 0\cdot(-1) + (-1)\cdot(-1) = 1 \\
(Q^{-1}AQ)_{21} &= 1\cdot 0 + (-1)\cdot 0 = 0 \\
(Q^{-1}AQ)_{22} &= 1\cdot(-1) + (-1)\cdot(-1) = 0
\end{aligned}
$$

$$
Q^{-1}AQ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}
$$

By part (i) the general solution is $\mathbf{x}(t) = (c_1 + c_2 t)\mathbf{v}_1 + c_2\mathbf{v}_2$:

$$x(t) = -c_1 - c_2 t + c_2, \qquad y(t) = -c_1 - c_2 t$$

Check: $\dot{x} = -c_2$ and $-x + y = -c_2$; $\dot{y} = -c_2$ and $-x + y$ again: both equations $\dot{x} = -x + y$, $\dot{y} = -x + y$ hold.

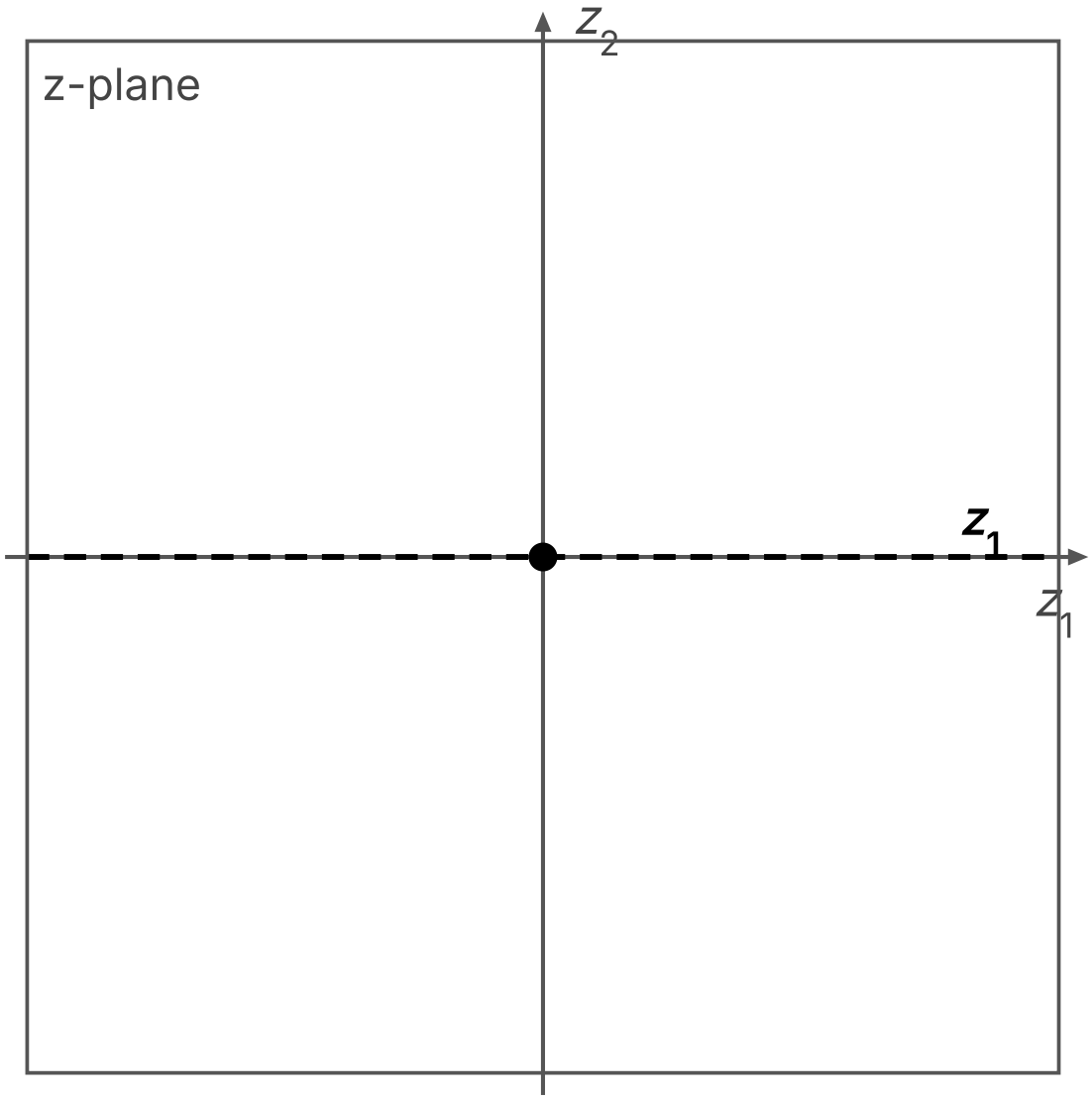
The coordinates are $\tilde{\mathbf{x}} = Q^{-1}\mathbf{x}$: $\tilde{x} = -y$, $\tilde{y} = x - y$ (first row $(0, -1)$, second row $(1, -1)$ of Q^{-1}). Every orbit is a straight line parallel to $\mathbf{v} = (1, 1)$; the line $y = x$ ($\tilde{y} = 0$) consists of equilibria.

Direction: $\dot{\mathbf{x}} = (y - x)(1, 1)^T$. Above the line $y = x$ ($y > x$) the motion is along $+(1, 1)$ (up and to the right); below it, along $-(1, 1)$ (down and to the left). The speed $|y - x|\sqrt{2}$ grows with the distance to the line of equilibria.

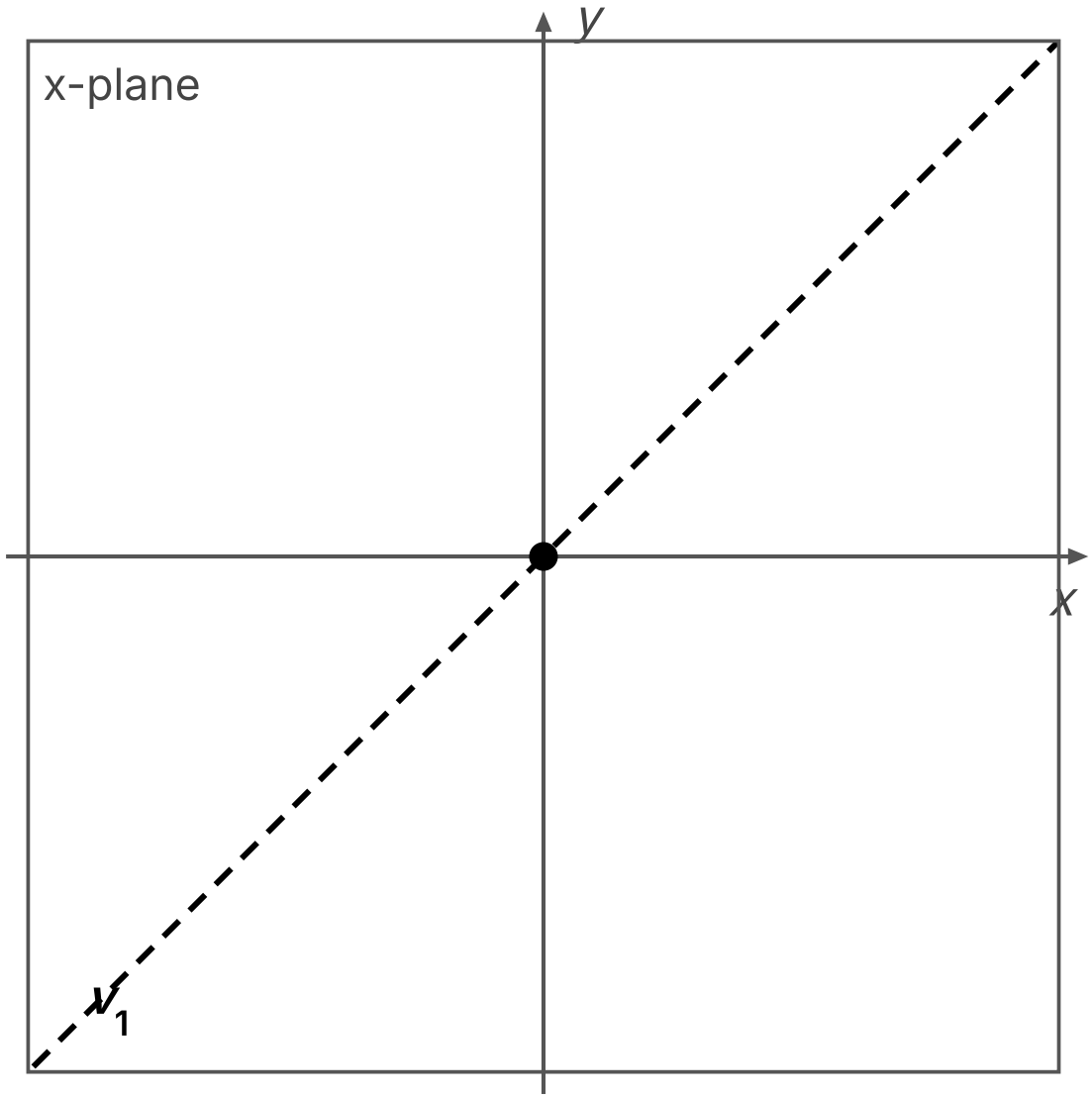
Phase portrait

$\lambda = 0$: every point of the eigenline through $\mathbf{v}_1 = (-1, -1)$ is an equilibrium. In the $\mathbf{z}$-plane $z_2 = c_2$ is constant and $z_1 = c_1 + c_2 t$ moves with constant speed c_2: horizontal lines, to the right if $z_2 > 0$ and to the left if $z_2 < 0$. Under $\mathbf{x} = Q\mathbf{z}$ they become lines parallel to $\mathbf{v}_1$: on the side of the eigenline where $\mathbf{v}_2$ points (for $z_2 > 0$) they are traversed along $+\mathbf{v}_1$, on the opposite side along $-\mathbf{v}_1$.

Orientation in the $\mathbf{x}$-plane, from $N\mathbf{x} = (A - \lambda I)\mathbf{x} = z_2\mathbf{v}_1$ (Topic 4 Sec. 8.9): at $(0, 1)^T$, $N(0, 1)^T = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -1\,\mathbf{v}_1$, so there the orbit drifts along $-\mathbf{v}_1$. The orbits are the straight lines parallel to $\mathbf{v}_1$.



Canonical $\mathbf{z}$-plane (line of equilibria (nilpotent shear)).



Original (x, y)-plane (line of equilibria (nilpotent shear)); dashed: the eigenline with $\mathbf{v}_1 = (-1, -1)$.

PHASE 4: INTERPRETATION AND CHECKS

- This is the limit $\lambda \to 0$ of the improper node $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$: the factor $e^{\lambda t}$ disappears and only the shear remains (the origin of the trace-determinant plane): the flow is $\mathbf{x}(t) = \mathbf{x}_0 + t\, A\mathbf{x}_0$ (the series of e^{At} stops after the linear term because $A^2 = O$): in (iii), $\mathbf{x}(t) = \mathbf{x}_0 + t(y_0 - x_0)(1, 1)^T$, which matches the direction and speed found.
- Check at a point: $\mathbf{x}_0 = (1, 0)$: formula gives $\mathbf{x}(t) = (1, 0) + t(0 - 1)(1, 1) = (1 - t, -t)$, and the general solution with $c_2 = \tilde{y}(0) = x_0 - y_0 = 1, c_1 = \tilde{x}(0) = -y_0 = 0$ gives $x = -(0 + t)(1) + 1 = 1 - t$ and $y = -t$. They coincide.
- $\operatorname{tr} A = \det A = 0$ is the origin of the trace-determinant plane (Topic 4 Sec. 9.1), where all the portrait types meet.

- The solutions are polynomials in t, so every orbit not on the line of equilibria leaves to infinity linearly in time in both directions.